

Arnav Gupta

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2020 - Present	B.Tech in Computer Science and Engineering	Indian Institute of Technology, Kanpur	9.5/10
2020	Class XII (CBSE)	Bal Bharati Public School, Dwarka	97.0%
2018	Class X (CBSE)	Bal Bharati Public School, Dwarka	96.6%

Scholastic Achievements

- Received **Academic Excellence Award** twice for exceptional performance in Academics at IIT Kanpur 2020, 2020-21
- Secured **All India Rank 174** in **JEE Advanced 2020** among more than 150,000 shortlisted candidates 2020
- Secured **All India Rank 285** in **JEE Mains 2020** among more than 1.2 million candidates 2020
- Qualified **KVPY(SX) 2020** with **All India Rank 264**, awarded National Research Fellowship 2020
- Qualified **KVPY(SA) 2019** with **All India Rank 521**, awarded National Research Fellowship 2019
- Cleared both stages of **NTSE 2018**, was among the 750 students selected for the NTS scholarship 2018
- Was among the 150 students in Delhi state who qualified **JSTSE 2017** 2017

Work Experience

Summer Intern | National Blockchain Project (CRUBN)

May'22 - Jul'22

- Worked on the **Data Integrity layer** in **SSI ecosystem** using **Hyperledger Fabric** and **Hyperledger Indy** blockchains
- Deployed mechanisms to securely fetch data for **Verifiable Credentials** from Fabric API into **ACA-Py Controllers**
- Issued **Universally Verifiable Credentials** on **Indy** verified using both **Indy** and **Fabric** with the help of **ZKPs**

Projects

Travel Allowance Reimbursement System 📄 | Course Project, Dr. Indranil Saha

Jan'22 - Apr'22

- Designed a portal to streamline the process of **Travel Allowance Reimbursement** for the IITK administration
- Implemented mechanisms of the app in the **Django** framework and hosted the project on **Heroku**
- Made Software Requirement Specification, Design, Implementation, Testing Documents and User Manual of the portal

Blocks and Chains 📄 | Association for Computing Activities (ACA), IIT Kanpur

May'21 - Jul'21

- Read literature on the basic mechanism, utilities, consensus mechanisms and applications of blockchains
- Programmed various core concepts (**Signature verification**, **hashing transaction data**, etc) of blockchains in **Javascript**
- Self-researched on **Smart Contracts** and **Game Theory** and made a smart contract for loan handling

Markowitz Portfolio Management 📄 | Finance & Analytics Club, IIT Kanpur

May'21 - Jul'21

- Analysed the risk and return of a sample of Indian stocks using libraries and **Modern Portfolio Theory**
- Plotted the **Efficient Frontier**— Obtained optimal weights to arrive at a portfolio having the Maximum Sharpe Ratio
- Calculated **Sortino Ratio**(1.91), **Treynor's Ratio**(11.63), **CAGR**(11.62) & **Beta**(0.75) and quantified **Risk&Reward**

Programming Experience

- Active participant in coding contests with maximum ratings of **1681** on **Codeforces** 📄 and **2042** on **Codechef** 📄
- Secured rank **306** out of more than 8000 participants in **Google Kickstart Round D 2022**
- Won the **Sport Programmer of the Month** - December Edition organised by Programming Club, IIT Kanpur

Relevant Coursework

Data Structures and Algorithms	Software Development and Operations	Computer Organisation
Probability Theory	Semantics and Logic	Mathematics for Computer Science
Fundamentals of Computing	Linear Algebra and ODE	Real Analysis

Technical Skills

Proficient: C, C++, Python, NodeJS

Utilities & frameworks: Git, Django, Tensorflow, Docker

Project Mentored

Machine Learning: Behind the curtain | Stamatics, IIT Kanpur

Mar'22 - Jun'22

- Mentored 80+ freshers and sophomores with a focus on the **mathematics** involved in **Machine Learning** algorithms
- Covered **Forward & Back propagation** (in Neural Networks), and **Gradient Descent** through programming assignments

Positions of Responsibility

Secretary | **Programming Club IIT Kanpur**

Aug'21 - May'22

- Conducted the Sport Programmer of the Month challenge for the batchmates in the month of December
- Set problems for programming contests in the college and contributed to various other events in this domain

Student Guide | **Counselling Service, IIT Kanpur**

Nov'21 - Jul'22

- Helped and counselled six freshmen to get acquainted with campus life and academics at IIT Kanpur